

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F
Q11	A	B	C	D	E	F
Q12	A	B	C	D	E	F
Q13	A	B	C	D	E	F
Q14	A	B	C	D	E	F

1. Which class of lever has the load situated in the middle?

- A. 1st class lever
- B. Reverse lever
- C. Compound lever
- D. 4th class lever
- E. 3rd class lever
- F. 2nd class lever

2. How long must the designed hand hold an item to meet the stated testable goals?

- A. 30 seconds
- B. 20 seconds
- C. 5 seconds
- D. 2 seconds
- E. 1 minute
- F. 10 seconds

3. Which term refers to how the anatomy of a real human hand influences the design?

- A. Societal Impact
- B. Accessibility & Cost
- C. Success Criteria Analysis
- D. Critical Failures
- E. Group Dynamics
- F. Bio-mimicry Connection

4. Where is the effort located in a 3rd class lever?

- A. In the middle
- B. Underneath the fulcrum
- C. Mixed with the load
- D. On the far left end only
- E. Behind the pivot point
- F. On the far right end only

5. What is the primary benefit of studying "bio-mimicry" (how the human hand works) before building the robot?

- A. To understand how nature creates movement and grip to solve engineering problems
- B. To prove that robots are stronger than humans
- C. To learn how to draw a hand for art class

- D. To see if human skin is waterproof
- E. To count how many bones are in the wrist
- F. To determine if the robot needs fingernails

6. Which anatomical component connects to the tendons and powers them?

- A. The bones
- B. The muscle
- C. The skin
- D. The fulcrum
- E. The wrist
- F. The joints

7. How is the "fulcrum" defined in relation to a lever?

- A. The weight of the load
- B. The place where the lever or levers are free to rotate
- C. The place where you apply force
- D. The result of mechanical advantage
- E. The rigid bar connecting the parts
- F. The object being lifted or moved

8. What key skill are students practicing when they assign roles, share responsibilities, and resolve disagreements?

- A. Collaboration and teamwork
- B. Advanced calculus
- C. Silent reading
- D. Memorization of facts
- E. Handwriting speed
- F. Competitive arguing

9. Why is it valuable for engineers to design low-cost robotic hands instead of only expensive ones?

- A. To avoid using modern materials
- B. To make the hands heavier and harder to use
- C. To stop people from using robotic hands
- D. To decrease the quality so the hand breaks faster
- E. To make helpful technology accessible to people who cannot afford thousands of dollars
- F. To make sure only wealthy people can use them

10. Why are robotic grippers necessary for industries like space exploration or handling hazardous waste?

- A. Because robots look cooler than humans in photos
- B. To use up extra electricity
- C. Because robots get lonely in space

D. To ensure the mission costs more money

E. To perform tasks that are too dangerous or deadly for humans to do directly

F. To prevent humans from having jobs

11. Why is the "Iteration & Redesign" step of the engineering process important for students?

A. To make the project take as long as possible

B. To copy exactly what the teacher built

C. To use up all the remaining glue and tape

D. To learn from failures and improve the performance of the design

E. To make the robotic hand look more colorful

F. To prove that the first attempt is always perfect

12. What component of the human hand acts like a string to pull the fingers toward the palm?

A. The fulcrum

B. The skin

C. The lever

D. The wrist

E. The tendons

F. The joints

13. Which part of the human hand provides enough friction so that gripped items do not slip away?

A. The wrist bone

B. The bones

C. The knuckles

D. The joints

E. The skin

F. The muscles

14. A lever is used to give the user which specific benefit?

A. Mechanical advantage

B. Increased friction

C. Thermal insulation

D. Reduced gravity

E. Electrical conductivity

F. A lighter load